

CONSOLE

The EFI console was designed to allow input from a wide variety of devices. This appendix provides examples of the mapping of keyboard input from various types of devices to EFI scan codes. While representative of common console devices in use today, it is not intended to be a comprehensive list. EFI application programmers can use this table to identify the EFI Scan Code generated by a specific key press. The description of the example device input data that generates a EFI Scan Code may be useful to EFI driver writers, as well as showing the limitations on which EFI Scan codes can be generated by different types of console input devices.

The EFI console was designed so that it could map to common console devices. This appendix explains how an EFI console could map to a VGA with PC AT 101/102, PC ANSI, or ANSI X3.64 consoles.

B.1 EFI_SIMPLE_TEXT_INPUT_PROTOCOL and EFI_SIMPLE_TEXT_INPUT_EX_PROTOCOL

Tables *EFI Scan Codes for EFI_SIMPLE_TEXT_INPUT_PROTOCOL* and *EFI Scan Codes for EFI_SIMPLE_TEXT_INPUT_EX_PROTOCOL* give examples of how input from a set of common input devices is mapped to EFI scan codes. Terminals and terminal emulators generally report function and editing keys as escape or control sequences. These sequences are formed by a control character followed by one or more additional graphic characters that indicate what the sequence means. ANSI X3.64 terminals generally require an ANSI parser to determine how to interpret a sequence and how to determine that the sequence is complete. These terminals can generate sequences using either 8-bit controls or 7-bit control sequences. Older terminal types, such as the VT100+ have a simpler set of sequences that can be interpreted using simple case statements. These terminals usually generate only 7-bit data, and 7-bit control sequences.

In the tables below, the CSI character is the 8-bit control character 0x9B, and is equivalent to the 7-bit control sequence “ESC [” (the 0x1B control ESC followed by the left bracket character 0x5B). The sequences are shown with spaces for readability, but do not contain the space character.

The VT100+ column represents a common class of terminal emulation that is a superset of the Digital Equipment Corporation (DEC) VT100 terminal. This includes VT-UTF8 (Hyperterm) and PC_ANSI terminal types. The ANSI X3.64 column shows the sequences generated by the DEC VT200 through VT500 terminals, which are an ANSI X3.64 / ISO 6429 compliant.

The USB HID and AT 101/102 columns show the scan codes generated by two common directly attached keyboards. These keyboards are generally used in combination with a VGA text display to form a “VGA Console”.

In the table below, the cells with N/A contained in them are simply intended to reflect that the key may be defined for that terminal or keyboard, but there is no industry standard or consistent mapping for the key. Some input devices might not implement all of these keys.

Table B.1: EFI Scan Codes for EFI_SIMPLE_TEXT_INPUT_PROTOCOL

EFI Scan Code	Description	ANSI DEC 500 mode)	X3.64 / VT200- (8-bit mode)	VT100+ (7-bit mode)	USB HID Values	Keyboard Values	AT Keyboard Codes	101/102 Scan Codes
0x00	Null scan code	N/A		N/A	0x00		N/A	
0x01	UP ARROW	CSI A		ESC [A	0x52		0xe0, 0x48	
0x02	DOWN ARROW	CSI B		ESC [B	0x51		0xe0, 0x50	
0x03	RIGHT ARROW	CSI C		ESC [C	0x4F		0xe0, 0x4d	
0x04	LEFT ARROW	CSI D		ESC [D	0x50		0xe0, 0x4b	
0x05	Home	CSI 1 ~		ESC h	0x4A		0xe0, 0x47	
0x06	End	CSI 4 ~		ESC k	0x4D		0xe0, 0x4f	
0x07	Insert	CSI 2 ~		ESC +	0x49		0xe0, 0x52	
0x08	Delete	CSI 3 ~		ESC -	0x4C		0xe0, 0x53	
0x09	Page Up	CSI 5 ~		ESC ?	0x4B		0xe0, 0x49	
0x0a	Page Down	CSI 6 ~		ESC /	0x4E		0xe0, 0x51	
0x0b	Function 1	CSI 1 1 ~		ESC 1	0x3A		0x3b	
0x0c	Function 2	CSI 1 2 ~		ESC 2	0x3B		0x3c	
0x0d	Function 3	CSI 1 3 ~		ESC 3	0x3C		0x3d	
0x0e	Function 4	CSI 1 4 ~		ESC 4	0x3D		0x3e	
0x0f	Function 5	CSI 1 5 ~		ESC 5	0x3E		0x3f	
0x10	Function 6	CSI 1 7 ~		ESC 6	0x3F		0x40	
0x11	Function 7	CSI 1 8 ~		ESC 7	0x40		0x41	
0x12	Function 8	CSI 1 9 ~		ESC 8	0x41		0x42	
0x13	Function 9	CSI 2 0 ~		ESC 9	0x42		0x43	
0x14	Function 10	CSI 2 1 ~		ESC 0	0x43		0x44	
0x17	Escape	ESC		ESC	0x29		0x01	

Table B.2: EFI Scan Codes for EFI_SIMPLE_TEXT_INPUT_EX_PROTOCOL

EFI Scan Code	Description	ANSI DEC 500 mode)	X3.64 / VT200- (8-bit mode)	VT100+ (7-bit mode)	USB HID Values	Keyboard Values	AT Keyboard Codes	101/102 Scan Codes
0x15	Function 11	CSI 2 3 ~		ESC !	0x44		0x57	
0x16	Function 12	CSI 2 4 ~		ESC @	0x45		0x58	
0x48	Pause	N/A		N/A	0x48		0xe1, 0x45	0x1d,
0x68	Function 13	CSI 2 5 ~		N/A	0x68		N/A	
0x69	Function 14	CSI 2 6 ~		N/A	0x69		N/A	
0x6A	Function 15	CSI 2 7 ~		N/A	0x6A		N/A	
0x6B	Function 16	CSI 2 8 ~		N/A	0x6B		N/A	
0x6C	Function 17	CSI 2 9 ~		N/A	0x6C		N/A	
0x6D	Function 18	CSI 3 0 ~		N/A	0x6D		N/A	
0x6E	Function 19	CSI 3 1 ~		N/A	0x6E		N/A	
0x6F	Function 20	CSI 3 2 ~		N/A	0x6F		N/A	
0x70	Function 21	N/A		N/A	0x70		N/A	
0x71	Function 22	N/A		N/A	0x71		N/A	
0x72	Function 23	N/A		N/A	0x72		N/A	
0x73	Function 24	N/A		N/A	0x73		N/A	

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0x7F	Mute	N/A	N/A	0x7F	N/A
0x80	Volume Up	N/A	N/A	0x80	N/A
0x81	Volume Down	N/A	N/A	0x81	N/A
0x100	Brightness Up	N/A	N/A	N/A	N/A
0x101	Brightness Down	N/A	N/A	N/A	N/A
0x102	Suspend	N/A	N/A	N/A	N/A
0x103	Hibernate	N/A	N/A	N/A	N/A
0x104	Toggle Display	N/A	N/A	N/A	N/A
0x105	Recovery	N/A	N/A	N/A	N/A
0x106	Eject	N/A	N/A	N/A	N/A
0x80-0xFFFF	OEM Reserved	N/A	N/A	N/A	N/A

B.2 EFI_SIMPLE_TEXT_OUTPUT_PROTOCOL for PC ANSI or ANSI X3.64 terminals

Table below defines how the programmatic methods of the EFI_SIMPLE_TEXT_OUTPUT_PROTOCOLs could be implemented as PC ANSI or ANSI X3.64 terminals. Detailed descriptions of PC ANSI and ANSI X3.64 escape sequences are as follows. The same type of operations can be supported via a PC AT type INT 10h interface.

Table B.3: Control Sequences to Implement EFI_SIMPLE_TEXT_INPUT_PROTOCOL

PC ANSI Codes	ANSI X3.64 Codes	Description
ESC [2 J	CSI 2 J	Clear Display Screen.
ESC [0 m	CSI 0 m	Normal Text.
ESC [1 m	CSI 1 m	Bright Text.
ESC [7 m	CSI 7 m	Reversed Text.
ESC [30 m	CSI 30 m	Black foreground, compliant with ISO Standard 6429.
ESC [31 m	CSI 31 m	Red foreground, compliant with ISO Standard 6429.
ESC [32 m	CSI 32 m	Green foreground, compliant with ISO Standard 6429.
ESC [33 m	CSI 33 m	Yellow foreground, compliant with ISO Standard 6429.
ESC [34 m	CSI 34 m	Blue foreground, compliant with ISO Standard 6429.
ESC [35 m	CSI 35 m	Magenta foreground, compliant with ISO Standard 6429.
ESC [36 m	CSI 36 m	Cyan foreground, compliant with ISO Standard 6429.
ESC [37 m	CSI 37 m	White foreground, compliant with ISO Standard 6429.
ESC [40 m	CSI 40 m	Black background, compliant with ISO Standard 6429.
ESC [41 m	CSI 41 m	Red background, compliant with ISO Standard 6429.
ESC [42 m	CSI 42 m	Green background, compliant with ISO Standard 6429.
ESC [43 m	CSI 43 m	Yellow background, compliant with ISO Standard 6429.
ESC [44 m	CSI 44 m	Blue background, compliant with ISO Standard 6429.
ESC [45 m	CSI 45 m	Magenta background, compliant with ISO Standard 6429.
ESC [46 m	CSI 46 m	Cyan background, compliant with ISO Standard 6429.
ESC [47 m	CSI 47 m	White background, compliant with ISO Standard 6429.
ESC [= 3 h	CSI = 3 h	Set Mode 80x25 color.

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ESC [row;col H	CSI row;col H	Set cursor position to row;col. Row and col are strings of ASCII digits.
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Table B.4: Example Keyboard Layout

Usage	EFI_KEY enum value	USB Keyboard HID Values	Typical AT-101 key position
Key a and A	EfiKeyC1	0x04	31
Key b and B	EfiKeyB5	0x05	50
Key c and C	EfiKeyB3	0x06	48
Key d and D	EfiKeyC3	0x07	33
Key e and E	EfiKeyD3	0x08	19
Key f and F	EfiKeyC4	0x09	34
Key g and G	EfiKeyC5	0x0A	35
Key h and H	EfiKeyC6	0x0B	36
Key i and I	EfiKeyD8	0x0C	24
Key j and J	EfiKeyC7	0x0D	37
Key k and K	EfiKeyC8	0x0E	38
Key l and L	EfiKeyC9	0x0F	39
Key m and M	EfiKeyB7	0x10	52
Key n and N	EfiKeyB6	0x11	51
Key o and O	EfiKeyD9	0x12	25
Key p and P	EfiKeyD10	0x13	26
Key q and Q	EfiKeyD1	0x14	17
Key r and R	EfiKeyD4	0x15	20
Key s and S	EfiKeyC2	0x16	32
Key t and T	EfiKeyD5	0x17	21
Key u and U	EfiKeyD7	0x18	23
Key v and V	EfiKeyB4	0x19	49
Key w and W	EfiKeyD2	0x1A	18
Key x and X	EfiKeyB2	0x1B	47
Key y and Y	EfiKeyD6	0x1C	22
Key z and Z	EfiKeyB1	0x1D	46
Key 1 and !	EfiKeyE1	0x1E	2
Key 2 and @	EfiKeyE2	0x1F	3
Key 3 and #	EfiKeyE3	0x20	4
Key 4 and \$	EfiKeyE4	0x21	5
Key 5 and %	EfiKeyE5	0x22	6
Key 6 and ^	EfiKeyE6	0x23	7
Key 7 and &	EfiKeyE7	0x24	8
Key 8 and *	EfiKeyE8	0x25	9
Key 9 and (	EfiKeyE9	0x26	10
Key 0 and)	EfiKeyE10	0x27	11
Enter*	EfiKeyEnter	0x28	43
Escape *	EfiKeyEsc	0x29	110
Del * / Backspace *	EfiKeyBackSpace	0x2A	15
Tab *	EfiKeyTab	0x2B	16
Spacebar	EfiKeySpaceBar	0x2C	61
Key - and _	EfiKeySpaceBar	0x2D	12
Key = and +	EfiKeySpaceBar	0x2E	13

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Key [and {	EfiKeySpaceBar	0x2F	27
Key] and }	EfiKeyD12	0x30	28
Key and	EfiKeyD13	0x31	29
Key ; and :	EfiKeyC10	0x33	40
Key ‘ and “	EfiKeyC11	0x34	41
Key ` and ~	EfiKeyE0	0x35	1
Key , and <	EfiKeyB8	0x36	53
Key . and >	EfiKeyB9	0x37	54
Key / and ?	EfiKeyB10	0x38	55
Capslock *	EfiKeyCapsLock	0x39	30
F1*	EfiKeyF1	0x3A	112
F2*	EfiKeyF2	0x3B	113
F3*	EfiKeyF3	0x3C	114
F4*	EfiKeyF4	0x3D	115
F5*	EfiKeyF5	0x3E	116
F6*	EfiKeyF6	0x3F	117
F7*	EfiKeyF7	0x40	118
F8*	EfiKeyF8	0x41	119
F9*	EfiKeyF9	0x42	120
F10*	EfiKeyF10	0x43	121
F11*	EfiKeyF11	0x44	122
F12*	EfiKeyF12	0x45	123
PrintScreen*	EfiKeyPrint	0x46	124
ScrollLock*	EfiKeySLck	0x47	125
Pause*	EfiKeyPause	0x48	126
Insert*	EfiKeyIns	0x49	75
Home*	EfiKeyHome	0x4A	80
PageUp*	EfiKeyPgUp	0x4B	85
Delete*	EfiKeyDel	0x4C	76
End*	EfiKeyEnd	0x4D	81
PageDown*	EfiKeyPgDn	0x4E	86
RightArrow*	EfiKeyRightArrow	0x4F	89
LeftArrow*	EfiKeyLeftArrow	0x50	79
DownArrow*	EfiKeyDownArrow	0x51	84
UpArrow*	EfiKeyUpArrow	0x52	83
NumLock*	EfiKeyNLck	0x53	90
Keypad /	EfiKeySlash	0x54	95
Keypad *	EfiKeyAsterisk	0x55	100
Keypad -	EfiKeyMinus	0x56	105
Keypad +	EfiKeyPlus	0x57	106
Keypad Enter*	EfiKeyEnter	0x58	108
Keypad 1 and End*	EfiKeyOne	0x59	93
Keypad 2 and DownArrow*	EfiKeyTwo	0x5A	98
Keypad 3 and	EfiKeyThree PageDown*	0x5B	103
Keypad 4 and LeftArrow*	EfiKeyFour	0x5C	92
Keypad 5	EfiKeyFive	0x5D	97
Keypad 6 and RightArrow*	EfiKeySix	0x5E	102
Keypad 7 and Home *	EfiKeySeven	0x5F	91
Keypad 8 and UpArrow*	EfiKeyEight	0x60	96
Keypad 9 and PageUp *	EfiKeyNine	0x61	101

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Keypad 0 and Insert *	EfiKeyZero	0x62	99
Keypad . and Delete *	EfiKeyPeriod	0x63	104
Menu	EfiKeyA4	0x76	n/a
LeftControl*	EfiKeyLCtrl	0xE0	58
LeftShift*	EfiKeyLShift	0xE1	44
LeftAlt*	EfiKeyLAlt	0xE2	60
LeftLogo*	EfiKeyA0	0xE3	127
RightControl*	EfiKeyRCtrl	0xE4	64
RightShift*	EfiKeyRShift	0xE5	57
RightAlt*	EfiKeyA2	0xE6	62
RightLogo*	EfiKeyA3	0xE7	128
Non-US # and ~	EfiKeyC12	0x32	42
Non-US and	EfiKeyIntl0	0x64	45
Non-US and _	EfiKeyIntl1	0x87	56
Non-US ¥ and	EfiKeyIntl3	0x89	n/a

An * indicates a non-printable character or keyboard behavior